Kids to School and Moms to Work:
New York City's Universal Pre-K Expansion and Mother's Employment

by

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Abstract

Using the restricted data from American Community Survey from 2011 to 2017, this paper examines the impact of New York City's (NYC) expansion of universal pre-kindergarten (UPK) on labor force participation of mothers with the youngest child of 4 years of age. Starting in Fall of 2014, any child who is 4 years old and residing in NYC for the past year is eligible for UPK for the academic year, for example all children born in 2010 would qualify for the academic year 2014-15. It uses a triple-difference approach - first compare mothers in NYC with the youngest child of 4-year-olds (treated mothers) to mothers with the youngest child of 5 and 6-year-olds (control mothers) before and after the program. Next, it compares this difference with mothers living in adjacent counties in the New York Metropolitan Area (NMA) in New York to NYC. I find that the program increased mothers' labor force participation by 5 percentage points (a 7.5 percent impact) in NYC. The results are robust to various robustness checks like comparing with mothers living in all of NMA and mothers in Philadelphia.

Keyword: Universal Pre-Kindergarten, Maternal Labor Force Participation, NYC

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1 Introduction

Since the 1970s, most developed countries have introduced government-subsidized preschool education[1]. Recently, publicly funded Universal Pre-Kindergarten (UPK) has gained significant attention in the United States. Advocates promote UPK as a tool to provide childcare and enhance school readiness, particularly for children from socioeconomically disadvantaged backgrounds (Magnuson et al., 2007; Weiland and Yoshikawa, 2013).

A growing body of research explores the long-term effects of early childhood education on outcomes such as health, education, and crime reduction in adulthood (Fitzpatrick, 2008, 2010; Herbst and Tekin, 2014; Hong et al., 2017; Gray-Lobe et al., 2021). Additionally, expanding UPK and other early childhood care programs may positively impact maternal labor force participation and working hours (Gelbach, 2002; Baker et al., 2008a; Bauernschuster and Schlotter, 2016; Malik, 2017; Ilin et al., 2022). However, findings on these labor market effects remain mixed.[2]

The 2014 expansion of New York City's Universal Pre-K (UPK) program provides a unique opportunity to examine the relationship between universal Pre-K availability and maternal employment. As the largest school district in the U.S., NYC offers an ideal setting for this analysis due to the scale and rapid implementation of the policy. Under NYC's education policy, any child born in 2010 and residing in the city became eligible for UPK in the 2014–2015 academic year—the program's inaugural year.[3] Using data from the American Community Survey (ACS) and a standard Difference-in-Differences framework, I compare the labor force participation rate of mothers whose youngest child is four years old—eligible for UPK—with those whose youngest child is five or six years old (already in grade school) before (academic years 2011–2013) and after (academic years 2014–2017) the program's implementation.[4]

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[1]The percentage of 3- to 5-year-olds enrolled in preschool has risen sharply in the last decade. For example, in Germany, Sweden, Norway, and the U.K., nearly 100% of children in this age group were enrolled in preschool by 2012, up from around 80% in 2002. While the OECD average stands at approximately 80%, the U.S. lags behind at around 65% (Cattan, 2016).

[2]For a review of the literature, see Morrissey (2017).

[3]https://insideschools.org/pre-kindergarten/how-to-enroll. Parents are required to provide proof of residency in NYC.

[4]Mothers whose youngest child was born in 2010 (i.e., mothers of four-year-olds) constitute the treatment group. In contrast, mothers whose youngest child was born in 2009 or 2008 (i.e., mothers of five- and six-year-olds, already in grade school) serve as the control group. For simplicity, I refer to academic years using

The identification of the above model relies on the assumption that, in the absence of UPK, the difference in labor force participation would have been similar among treated and control mothers. A key challenge, however, is that the UPK expansion coincided with a broader economic recovery in NYC, improving labor market conditions across all demographic groups, including mothers of four-year-olds. Consequently, the observed increase in maternal labor force participation between 2011 and 2017 may partly reflect macroeconomic trends rather than the causal impact of UPK. Additionally, differences in labor market outcomes between the treatment and control groups may be influenced by unobserved factors such as shifts in childcare availability and labor market conditions, potentially biasing the estimates.

To address this concern, I employ a triple-difference approach by comparing the Difference-in-Differences estimate from NYC with the same estimate for mothers living in neighboring counties to NYC within the New York Metropolitan Area (NMA)[5] of New York State that did not implement a universal Pre-K program. Given their geographic proximity and similar labor market conditions, these counties provide a reasonable counterfactual for NYC mothers. As an additional robustness check, I compare NYC mothers with those across the entire NMA and Philadelphia.[6] Using the triple-difference framework, I find that the labor force participation rate of NYC mothers with a UPK-eligible child increased by 0.05 percentage points (relative to a baseline of 0.67), a 7.5% increase. These results are robust to various sensitivity analyses and alternative specifications.

While the positive impact of UPK on children's development and maternal employment is widely acknowledged, its cost remains a major policy concern. Estimates suggest that nationwide implementation of universal preschool would cost approximately $34 billion per year, or about $6,600 per child (Gormley Jr et al., 2018). Other studies project annual costs as high as $351 billion, with per-child expenditures ranging from $12,500 to $15,000 (van Huizen et al., 2017). In New York City, UPK costs approximately $850 million annually. During the 2016–2017 school year, 72,176 students were enrolled across roughly 1,900 UPK programs, implying an average cost of $11,766 per student per year (of the New York

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the starting year (e.g., 2011–2012 as 2011).

[5]NMA consists of counties in NY, CT, and NJ.

[6]To my knowledge, no universal Pre-K program similar to NYC's existed in the NMA during this period. Given that the NMA is well connected to NYC via public transit, labor market conditions for mothers in both areas should be comparable.

State Comptroller, 2019).

If Universal Pre-K (UPK) significantly increases parental labor force participation and earnings, the resulting tax revenue gains could help offset program costs (Baker et al., 2008b). For instance, Humphries et al. (2024) find that UPK increases parents' earnings by as much as 21.7% per year, with effects persisting for at least six years. Similarly, Gormley Jr et al. (2018) estimate that the return on UPK investment is nearly three times its cost. A back-of-the-envelope calculation suggests that NYC's Pre-K expansion could generate up to $94 million annually in additional tax revenue, on top of other indirect long-term benefits such as increased educational attainment, labor productivity, and health benefits. Given the growing bipartisan interest in UPK, examining its impact on employment and tax revenue can provide valuable insights for policymakers (The Economist, 2019).

This study contributes to the growing literature on UPK's economic effects, particularly its impact on maternal employment. Previous research on Georgia and Oklahoma (Fitzpatrick, 2008; Li, 2020) has faced limitations due to inconsistencies in program implementation across school districts and small sample sizes. In contrast, NYC's rapid and universal rollout offers a unique setting to estimate the causal effects of UPK more precisely.

The rest of the paper is organized as follows: Section 2 provides background information; Section 3 discusses the data, variables, identification strategy, and methodology; Section 4 presents the results; Section 5 conducts robustness checks; Section 6 provides a cost-benefit analysis; and Section 7 concludes.

2 Background

In the past decade, the percentage of four-year-olds attending public pre-kindergarten (Pre-K) has increased sharply, nearly doubling to 33% (Malik, 2017). This phenomenon has had a dual impact: it not only improves children's preparedness for formal schooling—particularly for those from socioeconomically disadvantaged backgrounds—but also influences maternal labor force participation. In this paper, I focus on the latter aspect.

Research has consistently shown that mothers with young children are less likely to be hired and face significant wage penalties compared to fathers, a phenomenon often referred to as the "motherhood penalty" (Budig and England, 2002; Correll et al., 2007). Data from the U.S. Department of Labor reveal a stark contrast in labor force participation

rates for parents of young children: in 2015, mothers with children aged 3–5 had a labor force participation rate of approximately 67%, whereas fathers in the same cohort had a participation rate of 98% (of Labor Statistics, 2015). This sizable gap underscores the potential for public policy interventions, such as Universal Pre-K (UPK), to help close this divide.

A promising approach to addressing this gap is the expansion of UPK. Previous studies have explored the relationship between early childhood education programs and maternal labor force participation. Notably, from the mid-1960s to the late 1970s, several states introduced grants for school districts to implement kindergarten programs. Analyzing these early initiatives, Cascio (2009) found that for every ten children enrolled in public school as a result of these programs, approximately four mothers entered the labor force. More recently, several states and cities have expanded or introduced UPK programs. As of 2015, 42 states had some form of Pre-K subsidy, although not all programs are universally accessible (Cascio, 2017). Prominent examples of universal Pre-K programs include those in Oklahoma, Georgia, and Washington, D.C.

Although not directly comparable, internationally, several developed nations have implemented policies providing free preschool and childcare services. Research indicates that such policies have been successful in increasing maternal labor force participation. For example, countries such as Canada, Germany, Israel, Chile, and the United Kingdom have seen increases in maternal labor force participation rates of 5 to 10 percentage points as a result of free or subsidized preschool and childcare (Malik, 2017; Blau et al., 2003; Gornick and Meyers, 2007; Nollenberger and Rodríguez-Planas, 2015; Bousselin, 2022). However, the results are not universal. For example, Havnes and Mogstad (2011) find little to no effect on mothers' employment in Norway.

2.1 NYC – Pre-K program

New York City established a publicly financed Pre-K program for four-year-olds in 1998. However, due to a lack of funding and other political challenges, the program initially served only a small number of children. By 2013, NYC's Pre-K program enrolled 19,483 children in full-day programs and 39,045 children in part-day programs, accounting for just 18% and 37% of the total eligible population, respectively (Potter, 2015). In 2013, Bill de Blasio ran for election as mayor of NYC with universal Pre-K as a central part of his platform and won by a landslide. Following his election, the city began expanding Universal

Pre-K (UPK) in the fall of 2014. The expansion occurred in two phases: a portion of seats became available in 2014, and full expansion was implemented in 2015. According to official data, the city enrolled 53,000 children in September 2014 and 68,000 children in September 2015—nearly 100% of eligible children (Hong et al., 2017).

During the 2014 phase, the NYC Department of Education could not guarantee placements for all children. However, by 2015, in line with the policy's goal, all applicants were guaranteed placement, although not all were assigned to their first-choice locations (Hong et al., 2017). Notably, citing the success of UPK for four-year-olds, the city launched 3-K for All—targeting three-year-olds—in the fall of 2017.[7]

Recent research on similar UPK programs in urban areas has highlighted their potential to address educational inequities. For instance, studies of New York City's program have found positive effects on kindergarten readiness, particularly for children from lower-income families (Reardon and Robinson, 2018). Moreover, the expansion of UPK in large urban areas like New York City has the potential to increase maternal labor force participation—an important consideration in evaluating the broader economic impacts of these programs (Dearing and Zaslow, 2018).

3 Data and Variables, Identification Strategy and Methodology

3.1 Data

The primary source of data is the restricted American Community Survey (ACS) data from 2011 to 2017.[8] The ACS is a mixed-mode survey conducted on a rolling basis throughout the year and includes a rich set of household and individual characteristics. In this study, I primarily focus on self-reported labor force participation and other employment characteristics, along with various household characteristics.

[7] https://www1.nyc.gov/office-of-the-mayor/news/258-17/mayor-de-blasio-3-k-all#/0

[8] I obtained Special Sworn Status from the U.S. Census Bureau at the Baruch Research Data Center in 2018 in NYC.

3.2 Variables

The dependent variable is coded as 1 if a mother is in the labor force and 0 otherwise. Treatment mothers are defined as those whose youngest child is four years old and qualifies for UPK during a given academic year. Control group mothers are defined as those whose youngest child is five or six years old, as they are not impacted by the introduction of Universal Pre-K (UPK) for four-year-olds. The variable `Post` is coded as 1 for the academic years 2014/15–2016/17 and 0 for the academic years 2011/12–2013/14.

Additionally, mothers residing in New York City (NYC) (five boroughs) during the academic years are coded as 1, while mothers living in surrounding counties are coded as 0. Surrounding areas are defined in three ways: i) Mothers residing in the New York Metropolitan Area within New York State; ii) Mothers residing in counties that are entirely within the New York Metropolitan Area; and iii) Mothers residing in Philadelphia City, which is geographically close but outside the state.

Control variables include the number of children under 10 years old in the household (HH), single-parent households, the presence of grandparents in the household, household income, household size, mother's education level, mother's age, single motherhood status, and race (categorized as Non-Hispanic White, Black, Hispanic, and Other).

Insert Table 1 here

Table 1[9] presents summary statistics for all variables in New York City (NYC) and outside of NYC, comparing treated and control mothers before and after the program. As shown in the table, there are more single-parent households (HH) in NYC, a higher proportion of single mothers, and mothers tend to be younger in NYC compared to surrounding areas. Additionally, the racial composition in NYC includes a higher proportion of minority mothers.

[9]The number of observations is rounded off as required by the disclosure notice for all tables.

3.3 Identification Strategy and Estimating Equations

3.4 Difference in Differences Estimate (DD)

The main outcome of interest is the labor force participation of mothers whose youngest child is four years old in an academic year after 2014/15 while residing in New York City (NYC). I compare these mothers with those whose youngest child is five or six years old.

Thus, I analyze the difference in labor force participation between the treatment and control groups, as well as between the post- and pre-periods. This approach yields the standard difference-in-differences (DD or double difference) estimator. Using the definition of post and pre years as defined above, I estimate the following standard difference-in-differences regression model:

$$y_{it} = \beta_0 + \beta_1 Treat_{it} + \beta_2 Post_{it} + \beta_3 (Treat_{it} * Post_{it}) + \pi X_{it} + \delta_c + \lambda_t + \epsilon_{it} \qquad (1)$$

where y_{it} is the labor force participation status of a mother i in NYC in academic year t. $Treat_{it}$ indicates whether mother has a youngest child of 4-years-old, and $Post_{it}$ indicates whether mother is in pre or post period. X_{it} is a set of control variables; δ_c is a set of county dummies, λ_t is a vector of year dummies and ϵ_{it} is the error term and standard errors are clustered at the county level.

3.5 Triple Difference Estimate (DDD)

However, as discussed previously, the above estimation may suffer from several limitations. One of the key assumptions in this analysis is that, in the absence of the law, mothers in both the treatment and control groups would have exhibited similar labor force participation trends in both the pre- and post-periods. This assumption is known in the economics literature as the parallel trends assumption.

Intuitively, what if, before the start of the program, mothers in the treatment group already had higher labor force participation than those in the control group? If that were the case, my DD estimation might simply capture this pre-existing difference rather than the actual effect of the policy.

Second, the DD estimate could be confounded by omitted variables or unobserved events that differentially affected the labor force participation trends of mothers in NYC. For example, if labor market opportunities in NYC expanded between 2014 and 2017, any observed effect might reflect general labor market growth rather than the impact of the policy.

To account for NYC-specific, time-invariant factors common to both groups of mothers, I employ the triple-difference (DDD) method. Specifically, I compare the treatment and control groups with mothers living in neighboring counties around NYC. By using neighboring counties as a comparison group, I am able to control for broader regional trends that might have affected both groups similarly. I define a new variable, `NYC`, which takes a value of 1 for mothers residing in NYC and 0 for those living in neighboring counties (i.e., counties in the New York Metropolitan Area of New York State). Therefore, any remaining trends observed for either treated or control mothers in NYC must be NYC-specific and not due to broader labor market forces.

The DDD estimation of the impact of NYC's Universal Pre-K (UPK) expansion on the labor force participation of mothers with a four-year-old child is estimated as follows:

$$\begin{aligned} y_{int} = \beta_0 + \beta_1 Treat_i + \beta_2 Post_t + \beta_3 NYC_n + \beta_4 (NYC_n * Treat_i) \\ + \beta_5 (Treat_i * Post_t) + \beta_6 (NYC_n * Post_t) + \beta_7 (NYC_n * Post_t * Treat_i) \\ + \pi X_{it} + \delta_c + \lambda_t + \epsilon_{int} \end{aligned} \tag{2}$$

All variables are similar to above except NYC_n which 1 for mothers in NYC and 0 for mothers living in adjacent counties in NYC that are part of NMA. β_7 is the coefficient of interest.

4 Results

This section presents the results. Section 4.1 presents the main results, Section 4.2 tests the parallel trends assumption, Section 4.3 carries out an event study analysis, and Section 4.4 presents results for the Pre-K participation rate.

4.1 Main Results

Table 2 presents the difference-in-differences (DD) and triple-differences (DDD) estimates of the impact of Universal Pre-K (UPK) on mothers living in NYC before and after the program's implementation.

Panel A reports the DD estimates based on Equation (1). Only the coefficients β_1-β_3 are presented. Columns 1–4 display results with an increasing set of control variables: Column 1 presents results without controls. Column 2 adds county-level fixed effects. Column 3 incorporates both year and county-level fixed effects. Column 4 includes fixed effects along with the control variables mentioned earlier. Standard errors in all columns are clustered at the county level.

Panel B presents the main findings (only the coefficients β_1-β_7 are shown from Equation (2) above), showing that mothers whose youngest child was four years old were 5 percentage points more likely to participate in the labor force following the introduction of the UPK program in NYC—equivalent to an estimated 7.5 percent increase in labor force participation. Notably, the coefficients remain consistent as additional control variables are included, but the precision of the estimates improves.

Insert Table 2 here

One thing to note is that the coefficient on $Treat * Post$ is negative and significant, indicating that mothers with 4-year-olds living outside of NYC experienced a decline in labor force participation after 2014. This implies that NYC's UPK program helped mitigate the reduction in maternal labor force participation. Overall, these results suggest that the UPK program has a positive effect on mothers' labor force participation.

4.2 Parallel Trends

As discussed above, the validity of these estimates relies on the parallel trends assumption—meaning that, in the absence of the law, both groups of mothers in both types of locations would have exhibited similar labor force participation rates.

Since our data extends far enough back before the enactment of the UPK program, we can conduct a parallel trends test - a placebo test. To do so, we restrict the data to the period before 2014 and redefine the post and pre periods accordingly: Column 1: Post

equals 1 for the academic years 2012/13 and 2013/14 and 0 for 2011/12. Column 2: Post equals 1 for 2013/14 and 0 for 2012/13. Column 3: Post equals 1 for 2013/14, while 2011/12 and 2012/13 are coded as 0.

Across all columns, the results are statistically insignificant and close to zero, suggesting that there were no pre-existing differences in labor force participation trends between the treatment and control groups prior to UPK implementation. This supports the validity of the parallel trends assumption.

Insert Table 3 here

4.3 Event Study

Next, I conduct an event study design to track the evolution of UPK's impact over time. This approach allows us to observe how labor force participation trends evolved for both groups over time and to assess whether any pre-existing differences existed before the implementation of UPK.

Specifically, this involves estimating equation (2) above with year dummies interacted with the treatment status of mothers. First, I interact year with treatment status (mothers with a four-year-old vs. mothers with a five- or six-year-old) for those residing in NYC. Second, I interact year with treatment status for mothers living outside of NYC. I plot the coefficients in Figure 1 below.

Insert Figure 1 here

Figure 1 shows that the labor force participation rate for both groups of mothers was relatively stable around zero before the implementation of UPK. After the program's introduction, labor force participation increased for mothers with four-year-old children living in NYC.

4.4 First Stage Results

Did children's attendance in pre-K increase after the introduction of Universal Pre-K in NYC? If mothers are sending their children to pre-K and entering the labor force, then we would expect attendance to rise. Using publicly available ACS data from IPUMS, I con-

struct a dataset to measure whether children aged 3 to 5 are attending "nursery/preschool" programs.

In Figure 2 below, we observe that before the introduction of Universal Pre-K, the pre-K participation rate in NYC was lower than in areas outside the city. However, after the program's implementation, NYC begins to surpass those areas in participation.

Insert Figure 2 here

Next, I create a post variable equal to 1 for the years 2015–2017 and 0 for the years 2012–2014. I also identify whether children reside in NYC or outside of NYC, and run a difference-in-differences regression similar to Equation (1) above. Table 4 below shows the results. Following the introduction of the program, the percentage of children attending pre-K in NYC increased by 6 percentage points, a 9.5 percent impact.

Insert Table 4 here

5 Robustness Check

This section conducts robustness checks to assess the sensitivity of the results to data and control group selections. First, I compare mothers in New York City (NYC) with mothers living in the broader New York Metropolitan Area (NYM) and Philadelphia. Additionally, I compare mothers with children of different age groups. Next, given that the UPK program operates only during the academic year, I estimate the results after excluding the summer months. This adjustment ensures that the estimates focus on the program's impact during the academic year, avoiding potential bias from the summer months when children are not enrolled in the program. I also exclude the first year (2014/15) of the program, when participation rates were below 100% utilization. This exclusion helps mitigate any potential issues arising from incomplete program implementation in the first year. Finally, I compare the labor force participation rate between fathers and mothers to assess whether the program's impact is similar across genders.

5.1 Different comparision groups

First, I select different geographical control groups. Table 5, Column 1 compares mothers living in NYC with those in the New York Metropolitan Area (NYM). The results remain significant, though the estimates are slightly smaller. In Column 2, I compare mothers in NYC with those in Philadelphia, a city similar to NYC. The coefficient estimates are significant and slightly larger. Overall, the main findings remain robust across different comparison groups in the triple-difference model.

Second, I test the sensitivity of our model to different categorizations of treatment and control groups. In Column 3, I redefine treatment mothers. Instead of considering mothers whose youngest child is 5 or 6 years old as control mothers, I classify only those with a youngest child of 5 as controls. The results remain significant, with slightly larger estimates. In Column 4, I code mothers with a 5-year-old as 1 and those with a 6-year-old as 0. The results are not significant, indicating that defining control mothers as those with a 5-year-old and 6-year-old is not sensitive to the main findings above.

Finally, since the UPK program operates only during the academic year, I exclude the summer months (June, July, and August) from the analysis in Column 5. The coefficients remain significant, with an increase in magnitude. Additionally, in Column 6, I exclude the 2014/15 academic year, the program's first year when participation was below 100%. The results remain very similar to the main estimates.

Insert Table 5 here

The results are significant across different comparison groups (geography and age groups).

5.2 Comparing with Fathers' Labor Force

Table 6, Column 1 compares the labor force characteristics of fathers. Specifically, I repeat Equations (1) and (2) above, but for fathers instead of mothers. The results indicate no significant difference in labor force characteristics, suggesting that the UPK program has no impact on fathers' labor force participation, underscoring the gendered nature of caregiving responsibilities. In Column 2, I compare mothers and fathers in NYC. The treatment is coded as 1 for mothers and 0 for fathers. The results show that the labor force participation

rate of mothers with a youngest child of four years old increased by 6 percentage points compared to fathers.

These findings might indicate that mothers in the treatment group experienced a notable increase in labor force participation relative to fathers, supporting the idea of the motherhood penalty. The lack of supportive family policies, such as paid parental leave and subsidized childcare, exacerbates these penalties, particularly in the United States (Blau and Kahn, 2013).

Insert Table 6 here

6 Cost-Benefit Analysis

As mentioned above, the cost of Universal Pre-K (UPK) is very high. NYC spends approximately $850 million annually on UPK. During the 2016–2017 school year, 72,176 students were enrolled across roughly 1,900 UPK programs, implying an average cost of $11,766 per student per year (of the New York State Comptroller, 2019). Universal Pre-K in New York City has a measurable impact on maternal labor force participation, potentially leading to a significant increase in tax revenue at the federal, state, and local levels, thereby offsetting some of the program's costs.

This study finds that UPK increased the maternal labor force participation rate in NYC by 7.5 percent. On average, approximately 50,000 mothers in NYC have four-year-old children eligible for UPK enrollment.[10] Given this, an estimated 3,750 additional mothers enter the workforce annually due to the availability of UPK. The increase in employment among these mothers generates substantial fiscal benefits, as they contribute to federal, state, and local income taxes, as well as payroll taxes. Below, I provide a simple back-of-the-envelope calculation to quantify these effects. Based on U.S. Census Bureau data, the median annual income for women in this demographic in NYC is $70,663, which forms the basis for tax revenue calculations (Bureau, 2017).[11]

[10] According to the authors' calculations from the ACS data used for this study.

[11] These estimates rely on several key assumptions. First, the 7.5 percent labor force participation increase is based solely on this study, which measures the impact of UPK availability on maternal employment. The total number of eligible mothers with four-year-olds is derived from NYC ACS data, assuming that

At the federal level, income tax revenue generated from these newly employed mothers is estimated using an effective tax rate of 17 percent, resulting in approximately $45.05 million in additional federal tax revenue annually. At the state level, New York State collects income tax at an estimated effective rate of 6.33 percent, yielding an additional $17.57 million per year. The City of New York also benefits from local income taxes, applied at an approximate rate of 3.5 percent, contributing $9.75 million annually. Beyond these income taxes, payroll taxes, including Social Security and Medicare contributions, amount to 7.65 percent of earnings, generating an additional $20.27 million in revenue. When combined, the total additional tax revenue attributed to increased maternal employment facilitated by UPK reaches approximately $92.64 million per year.

While this fiscal analysis demonstrates that UPK generates significant tax revenue gains, its broader policy implications extend beyond immediate government revenues. Increased maternal labor force participation contributes to long-term career progression and higher lifetime earnings, resulting in sustained economic benefits. Additionally, reductions in reliance on public assistance programs—including food stamps and childcare subsidies—generate further cost savings for the government. From an educational perspective, research suggests that children who attend Pre-K programs achieve better academic outcomes, ultimately leading to higher future earnings and increased tax contributions (Heckman, 2011). Other long-term societal benefits include lower crime rates and greater social mobility, both of which contribute to a more productive workforce and reduced government spending on incarceration and remedial education programs (Bartik et al., 2012, 2023).

Overall, the estimated $92.64 million in annual tax revenue represents a substantial fiscal return on investment for the government. When additional economic and social benefits are considered, the long-term returns of Universal Pre-K likely exceed its immediate costs, reinforcing the case for continued program expansion. Future research should incorporate dynamic modeling approaches to account for wage growth, evolving labor force participation trends, and broader societal benefits that extend beyond tax revenue alone.

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participation rates remain consistent across years. The median annual income of $70,663 is based on 2017 census figures and is assumed to be representative of the earnings potential for newly employed mothers. Tax rate calculations reflect effective rates for this income level, assuming standard deductions and joint filing status where applicable. Additionally, the analysis does not incorporate potential wage growth over time, meaning that actual tax contributions may increase in future years as earnings rise.

7 Discussion and Conclusion

Although the primary objective of Universal Pre-K (UPK) is to enhance children's school readiness, this paper demonstrates that it also yields significant unintended benefits by increasing maternal labor force participation. Using a triple-difference approach, I find that NYC's UPK expansion led to a 7.5% increase in the labor force participation rate of mothers with an eligible child. These findings provide compelling evidence that access to free preschool services alleviates childcare constraints and strengthens workforce attachment. The results remain robust across multiple specifications, sensitivity analyses, and alternative control groups, reinforcing the reliability of the estimated effects.

The policy implications of these findings are substantial. With growing bipartisan support for the expansion of UPK programs (The Economist, 2019), understanding their broader economic impact is essential for policymakers tasked with allocating public resources efficiently. Since these programs are primarily financed through public funds, a critical question is whether the economic returns generated by increased maternal employment justify the costs of UPK provision. A back-of-the-envelope calculation suggests that the increased maternal labor force participation translates into approximately $93 million in additional annual tax revenue. This figure likely represents a conservative estimate, as existing research indicates that labor market gains from childcare access tend to persist over time. The long-term fiscal benefits of UPK investments may thus be even larger, as mothers who re-enter the workforce or increase their working hours experience continued earnings growth, enhanced career stability, and greater retirement savings.

Despite these promising findings, important questions remain. A key consideration for policymakers is how the return on investment in UPK compares to other public expenditures, particularly in the context of early childhood education and labor market policies. Given the limitations of survey data, future research should leverage administrative records and longitudinal datasets to conduct a more comprehensive cost-benefit analysis. Such an analysis would be particularly valuable for the NYC government and other policy stakeholders seeking to optimize early childhood education investments. Additionally, further research should examine the long-term effects of UPK-induced labor force participation on mothers' career trajectories, wage progression, and household financial stability.

Importantly, the program's effects are likely heterogeneous, particularly among minority, low-income, and immigrant mothers, who may face greater structural barriers to workforce participation due to childcare. However, due to restrictions on the dataset

used in this study, a more granular analysis to explore these differential impacts does not pass disclosure requirements. Future research with expanded data access should investigate these heterogeneous effects in greater detail to inform more targeted policy interventions. Understanding which groups benefit the most—or remain underserved—can help refine UPK policies to maximize their impact on economic mobility and equity.

As cities and states across the country consider expanding publicly funded preschool programs, these findings underscore UPK's dual role as both an educational intervention and a labor market policy. By improving early childhood learning outcomes while simultaneously enhancing maternal employment prospects, UPK represents a policy with far-reaching economic and social benefits. Its ability to support working families, strengthen local economies, and promote gender equity in the workforce makes it a compelling area for continued policy attention, rigorous evaluation, and sustained investment.

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Tables

Table 1: Summary Statistics

		NYC		Outside NYC	
		Treat	Control	Treat	Control
In Laborforce	Post	0.70	0.71	0.69	0.74
	Pre	0.67	0.71	0.71	0.73
Children <10 yrs in HH	Post	0.34	0.42	0.31	0.45
	Pre	0.39	0.43	0.33	0.43
Single Family HH	Post	0.23	0.19	0.17	0.15
	Pre	0.21	0.18	0.19	0.13
Grandparents in HH	Post	0.05	0.07	0.04	0.04
	Pre	0.06	0.05	0.05	0.04
HH income (000s)	Post	110	110	150	150
	Pre	100	100	140	140
HH size	Post	4.41	4.39	4.62	4.54
	Pre	4.45	4.29	4.74	4.49
Mothers Edu	Post	18.03	17.95	19.13	19.10
	Pre	17.97	17.79	19.16	19.07
Mothers Age	Post	34.99	37.01	36.82	38.64
	Pre	34.93	36.59	36.88	38.52
Non-Hispanic White	Post	0.26	0.24	0.55	0.56
	Pre	0.25	0.23	0.56	0.55
Non-Hispanic Black	Post	0.22	0.22	0.09	0.11
	Pre	0.26	0.24	0.09	0.12
Hispanic	Post	0.38	0.37	0.26	0.23
	Pre	0.34	0.39	0.24	0.24
Others	Post	0.15	0.17	0.10	0.10
	Pre	0.15	0.13	0.11	0.10
Single Mothers	Post	0.39	0.39	0.19	0.23
	Pre	0.40	0.41	0.23	0.21
Obs		3800	6900	2400	4600

Notes: Treat equals mothers with youngest children of 4 years old. Control equals mothers with youngest children of 5 and 6 years old. NYC equals mothers living in NYC and Outside NYC equals mothers living in counties outside NYC in New York State.

Table 2: Impact of Universal Pre-k on mothers' labor force participation

Dependent Variable: Mothers' in Laborfocre				
Panel A: Double Difference				
	(1)	(2)	(3)	(4)
PostxTreat	0.024	0.022	0.024*	0.024**
	(0.012)	(0.011)	(0.011)	(0.008)
Post	0.001	0.002	0.017	0.018
	(0.014)	(0.014)	(0.032)	(0.033)
Treat	-0.038**	-0.038**	-0.040**	-0.040**
	(0.011)	(0.011)	(0.011)	(0.012)
N	11000	11000	11000	11000
R-squared	0.001	0.005	0.006	0.080
Dep Var Mean	0.702	0.702	0.702	0.702
Panel B: Tripple Difference				
NYCxPostxTreat	0.050**	0.048**	0.050**	0.050***
	(0.020)	(0.020)	(0.020)	(0.015)
NYCXPost	-0.006	-0.005	-0.005	-0.002
	(0.021)	(0.022)	(0.021)	(0.020)
TreatxPost	-0.026	-0.026	-0.027	-0.028**
	(0.017)	(0.017)	(0.017)	(0.013)
NYCXTreat	-0.020	-0.020	-0.020	-0.027
	(0.022)	(0.022)	(0.022)	(0.021)
NYC	-0.018	-0.000	-0.000	0.014
	(0.019)	(0.010)	(0.010)	(0.015)
Post	0.008	0.007	0.024	0.024
	(0.017)	(0.017)	(0.019)	(0.018)
Treat	-0.019	-0.019	-0.018	-0.014
	(0.020)	(0.020)	(0.020)	(0.017)
N	18000	18000	18000	18000
R-squared	0.002	0.005	0.005	0.077
Dep Var Mean	0.709	0.709	0.709	0.709
Fixex Effects	No	Counties	Year/Cnty	Year/Cnty
Controls	No	No	No	Yes

Notes: Panel A reports the coefficients of DD regressions and Panel B reports the DDD regression coefficients. All regressions control for mother's education, mother's age, mother's race (Non-Hispanic White, Non-Hispanic Black, Hispanic and Others), household income, household size, if siblings older than 10 years are present, and grandparents are living in household, and whether the mother is single (rather than married or cohabiting). It also controls for year and county fixed effects when indicated. All estimates are weighted at person level. Standard errors are clustered at the county level and are in parenthesis. The symbols *, ** and *** represent significance at the 10, 5 and 1 percent, respectively.

Table 3: Pre-K Participation Rate

PostxTreat	0.058**
	(0.021)
Post	-0.022
	(0.026)
Treat	-0.191***
	(0.010)
N	16387
R-squared	0.016
Dep Var Mean	0.605

Notes: Only the main coefficients are reported. Post = 1 for the years 2015-2017 and 0 for the year 2012-2014. All regressions control for mother's education, mother's age, mother's race (Non-Hispanic White, Non-Hispanic Black, Hispanic and Others), household income, household size, if siblings older than 10 years are present, and grandparents are living in household, and whether the mother is single (rather than married or cohabiting). It also controls for year and county fixed effects when indicated. All estimates are weighted at person level. Standard errors are clustered at the county level and are in parenthesis. The symbols *, ** and *** represent significance at the 10, 5 and 1 percent, respectively.

Table 4: Parallel Trend Test

Dependent Variable: Mothers' in Labor force			
Panel A: Double Difference			
	(1)	(2)	(3)
PostxTreat	-0.025	0.011	-0.006
	(0.030)	(0.032)	(0.038)
Post	0.026	0.028	0.019
	(0.025)	(0.031)	(0.030)
Treat	-0.025	-0.053**	-0.039
	(0.024)	(0.016)	(0.019)
N	6000	3700	5800
R-squared	0.081	0.074	0.081
Dep Var Mean	0.697	0.697	0.697
Panel B: Tripple Difference			
NYCxPostxTreat	-0.019	0.053	0.026
	(0.038)	(0.048)	(0.052)
NYCXPost	-0.010	0.040	0.026
	(0.028)	(0.030)	(0.030)
TreatxPost	-0.004	-0.042	-0.033
	(0.025)	(0.037)	(0.037)
NYCXTreat	-0.013	-0.060***	-0.033
	(0.031)	(0.018)	(0.022)
NYC	0.043*	0.001	0.029
	(0.020)	(0.031)	(0.021)
Post	0.025	-0.013	0.002
	(0.020)	(0.014)	(0.018)
Treat	-0.014	0.006	-0.007
	(0.021)	(0.010)	(0.014)
N	10000	6100	9500
R-squared	0.074	0.073	0.075
Dep Var Mean	0.706	0.709	0.706

Notes: Panel A reports the coefficients of DD regressions and Panel B reports the DDD regression coefficients. Column 1, Post = 1 for the years 2013 and 2014 and 0 for the year 2012. Column 2 Post =1 for 2014 and 0 for 2013. Column 3 Post = 1 for 2014 and 0 for 2012 and 2013. All regressions control for mother's education, mother's age, mother's race (Non-Hispanic White, Non-Hispanic Black, Hispanic and Others), household income, household size, if siblings older than 10 years are present, and grandparents are living in household, and whether the mother is single (rather than married or cohabiting). It also controls for year and county fixed effects when indicated. All estimates are weighted at person level. Standard errors are clustered at the county level and are in parenthesis. The symbols *, ** and *** represent significance at the 10, 5 and 1 percent, respectively.

Table 5: Different Comparison Groups and age cutoffs

	NYC v NMA (1)	NYC v Philly (2)	Age 4 v 5 (3)	Age 5 v 6 (4)	Drop Summer (5)	Drop 2014/15 (6)
Panel A: Double Difference						
PostxTreat	0.024**	0.024**	0.043**	0.027	0.053**	0.028*
	(0.008)	(0.008)	(0.014)	(0.019)	(0.017)	(0.011)
Post	0.018	0.018	0.008	0.018	0.008	0.031
	(0.033)	(0.033)	(0.021)	(0.028)	(0.035)	(0.041)
Treat	-0.040**	-0.040**	-0.043	-0.024	-0.041**	-0.044**
	(0.012)	(0.012)	(0.023)	(0.013)	(0.011)	(0.011)
N	11000	11000	7000	6000	9000	9000
R-squared	0.080	0.080	0.081	0.080	0.084	0.083
Dep Var Mean	0.702	0.702	0.693	0.731	0.702	0.700
Panel B: Tripple Difference						
NYCxPostxTreat	0.036**	0.086***	0.072**	0.031	0.082***	0.053***
	(0.014)	(0.008)	(0.031)	(0.049)	(0.019)	(0.015)
NYCXPost	-0.005	-0.061***	-0.024	0.010	-0.017	0.003
	(0.016)	(0.013)	(0.023)	(0.025)	(0.023)	(0.018)
TreatxPost	-0.013	-0.062***	-0.031	-0.003	-0.030***	-0.026**
	(0.011)	(0.003)	(0.027)	(0.046)	(0.008)	(0.011)
NYCXTreat	-0.019	-0.086***	-0.037	-0.013	-0.026	-0.027
	(0.015)	(0.011)	(0.034)	(0.020)	(0.020)	(0.021)
NYC	-0.066***	0.019*	0.034*	-0.000	0.023	0.021
	(0.009)	(0.008)	(0.017)	(0.013)	(0.015)	(0.016)
Post	0.021	0.076***	0.035	-0.003	0.028	0.026
	(0.013)	(0.017)	(0.026)	(0.022)	(0.020)	(0.017)
Treat	-0.024**	0.045***	-0.006	-0.013	-0.015	-0.020
	(0.011)	(0.002)	(0.025)	(0.017)	(0.017)	(0.018)
N	30000	12000	12000	11000	15000	14000
R-squared	0.065	0.084	0.080	0.067	0.079	0.078
Dep Var Mean	0.719	0.713	0.700	0.741	0.710	0.706

Notes: Panel A reports the coefficients of DD regressions and Panel B reports the DDD regression coefficients. Column 1 compares mothers living in NYC with mothers living in New York Metropolitan Area, column 2 with mothers living in Philadelphia City. Column 3 compares mothers with youngest child of 4-year-old with 5-year-old, column 4 compares 5-year-old to 6-year-old. Column 5 drops observations from June, July and August and column 6 drops observations for the academic year 2014 and 2015. All regressions control for mother's education, mother's age, mother's race (Non-Hispanic White, Non-Hispanic Black, Hispanic and Others), household income, household size, if siblings older than 10 years are present, and grandparents are living in household, and whether the mother is single (rather than married or cohabiting). It also controls for year and county fixed effects when indicated. All estimates are weighted at person level. Standard errors are clustered at the county level and are in parenthesis. The symbols *, ** and *** represent significance at the 10, 5 and 1 percent, respectively.

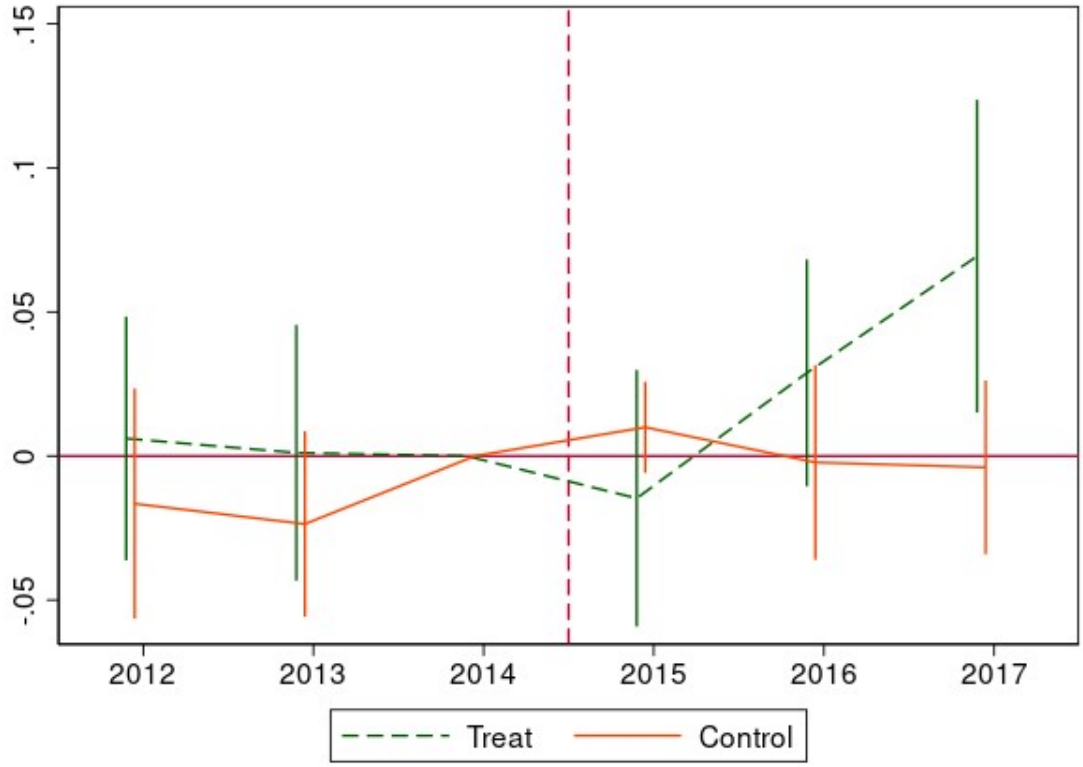
Table 6: Mothers vs Fathers

	Fathers Only (1)	Mother v Father (2)
Panel A: Double Difference		
PostxTreat	-0.003	0.037**
	(0.007)	(0.013)
Post	-0.008	0.004
	(0.012)	(0.016)
Treat	0.018***	-0.299***
	(0.003)	(0.028)
N	7000	6000
R-squared	0.029	0.144
Dep Var Mean	0.935	0.788
Panel B: Tripple Difference		
NYCxPostxTreat	-0.009	0.060**
	(0.013)	(0.021)
NYCXPost	-0.006	-0.012
	(0.009)	(0.016)
TreatxPost	0.005	-0.024
	(0.012)	(0.017)
NYCXTreat	0.016**	-0.043
	(0.006)	(0.030)
NYC	-0.021**	0.019
	(0.007)	(0.013)
Post	-0.005	0.017
	(0.008)	(0.013)
Treat	0.002	-0.262***
	(0.005)	(0.018)
N	13000	11000
R-squared	0.028	0.147
Dep Var Mean	0.945	0.800

Notes: Panel A reports the coefficients of DD regressions and Panel B reports the DDD regression coefficients. Column 1 compares fathers with youngest child of 4-year-old with 5 and 6 years old and column 2 compares mothers with youngest child of 4-year-old with fathers with youngest child of 4-year-old. All regressions control for mother's education, mother's age, mother's race (Non-Hispanic White, Non-Hispanic Black, Hispanic and Others), household income, household size, if siblings older than 10 years are present, and grandparents are living in household, and whether the mother is single (rather than married or cohabiting). It also controls for year and county fixed effects when indicated. All estimates are weighted at person level. Standard errors are clustered at the county level and are in parenthesis. The symbols *, ** and *** represent significance at the 10, 5 and 1 percent, respectively.

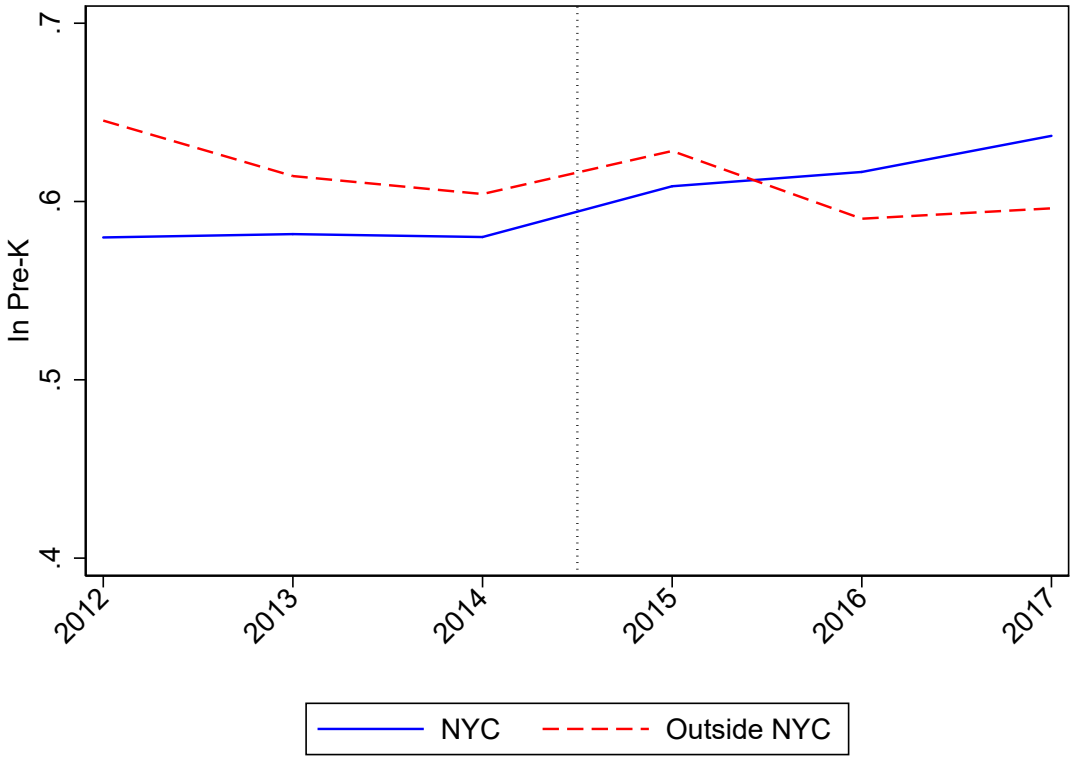
Figures

Figure 1: Event Study Graph



Source: Author's calculation. Horizontal bars are 95% CI. 2013/2014 is realtive academic year. Treat refers to mothers living in NYC and control refers to mothers living outside of NYC.

Figure 2: Pre-K Attendence



Source: Author's calculation using IPUMS ACS data.